

Ontology as a Hyperparameter: Task-Aware Knowledge Graph Construction Using LLMs

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Abstract

This project investigates the construction of task-aware knowledge graphs (KGs) from text, treating graph structure as an optimization target rather than a fixed representation. Existing frameworks such as Neo4j-based pipelines and LLM-driven tools (e.g., LlamaIndex) focus on constructing large, general-purpose graphs without incorporating feedback from downstream tasks, despite evidence from evaluation frameworks (e.g., KGrEaT, GEval, KGBench) that KG utility is task-dependent. The research addresses this gap by proposing a feedback-driven approach in which ontology is treated as a tunable component influencing predictive performance. The study uses document-based datasets linking text to measurable outcomes, including financial news (FNSPID) paired with stock price movements. LLM agents are employed for ontology evolution and triple extraction, enabling scalable generation of multiple KG variants. Graph-derived features are then computed over temporal windows and used in predictive models, whose performance is fed back to the agents to refine ontology design, closing the loop between graph construction and downstream task. Preliminary results indicate that graph structure strongly influences predictive outcomes, with adaptive ontology updates leading to significant improvements.

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1 Introduction

Representing knowledge in structured forms that intelligent systems can query and reason over is a foundational challenge of artificial intelligence. Knowledge graphs (KGs) have become the attractive medium for this purpose, encoding entities, relationships, and semantic descriptions extracted from text to support applications from question answering and recommendation to predictive modelling Ji et al. (2021). The ability to construct KGs from unstructured text at scale has been substantially advanced by large language models (LLMs), which enable flexible extraction of entities and relations without hand-crafted rules Mo et al. (2025). Despite this progress, a fundamental tension persists in KG construction: ontology and graph structure are typically specified *a priori* and evaluated independently of the tasks for which the graph is ultimately used.

Existing KG construction tools — from manual ontology engineering to LLM-driven frameworks such as LlamaIndex and LangChain — treat the ontology as a fixed design choice established before construction begins and unchanged throughout. Evaluation of the resulting graph is typically conducted through structural quality metrics: completeness, consistency, and redundancy Paulheim (2017). Paulheim (2024) draw a sharp distinction between this intrinsic mode of evaluation and task-based evaluation — assessing a KG by its utility on a concrete downstream application. Frameworks such as KGrEaT Heist et al. (2023) and KGBench Bloem et al. (2021) try to formalise the approach to such evaluations. As Heist et al. (2023) observe: *“Performance indicators may be used to make an informed decision when picking a KG for a given task. . . and to compare different versions of a KG during its life cycle.”* Yet despite this insight, no existing work closes the loop at construction time: no downstream evaluation signal has been used to directly steer KG ontology evolution during the building process itself

This paper aims to address the gap between graph construction and downstream task performance by introducing a feedback-driven framework in which the KG ontology is treated as a tunable hyperparameter. Rather than fixing the schema in advance, we propose a closed-loop system:

- (i) An Ontology Agent proposes candidate ontology configurations.
- (ii) A KG Builder Agent constructs knowledge graphs from text under each configuration.
- (iii) Graph-derived features are computed and evaluated on a downstream task.
- (iv) The resulting task performance score is returned as a feedback signal to guide the next ontology revision.

This loop repeats until performance converges.

We instantiate this framework on the task of next-day stock price movement prediction using the FNSPID financial news dataset Dong et al. (2024), a corpus of financial news articles linked to NASDAQ price time series. This domain offers a clear and measurable downstream objective, dense entity interactions, and a natural temporal structure that motivates subgraph-based feature extraction. The framework is designed to be domain-agnostic and is extensible to other settings.

Contributions. This paper makes the following contributions:

1. We propose a feedback-driven KG construction framework that treats ontology as a hyperparameter optimized for downstream task performance, closing the loop between graph construction and task evaluation.
2. We introduce a structured feature representation for temporal KG subgraphs — comprising node type histograms, relation type histograms, metapath counts, temporal novelty statistics, and topology features — that supports lightweight downstream task evaluation without requiring graph neural networks.
3. We provide empirical evidence on the FNSPID dataset that task-aware ontology adaptation improves downstream task performance, enables more compact graph representations without sacrificing performance, and produces more informative graph structures over successive iterations.

Paper organisation. Section 2 reviews related work on KG construction, task-based KG evaluation, and task-aware graph learning. Section 3 states the research hypotheses formally. Section 4 describes the proposed framework in detail. Section 5 presents the experimental setup, and Section 6 reports results. Section 7 discusses findings and limitations, and Section 8 concludes.

2 Related Work

The advent of large language models has substantially lowered the barrier to KG construction at scale, and the popularity of this approach has led to a proliferation of tools and frameworks for LLM-augmented KG construction. Pan et al. (2024) survey this landscape, classifying LLM-augmented construction methods by the role of the LLM — entity discovery, relation extraction, end-to-end generation, or knowledge distillation. In knowledge graph building three dominant open source providers are LlamaIndex, LangChain, and Neo4j. LlamaIndex (Liu, 2022) offers three extraction strategies for property graph construction (LlamaIndex, 2024): a schema-free extractor in which the LLM assigns entity and relation types freely (`SimpleLLMPathExtractor`); a strict schema extractor that

enforces a user-defined list of allowed types (`SchemaLLMPathExtractor`); and a guided extractor that accepts an initial type list as non-binding hints while permitting the LLM to introduce novel types (`DynamicLLMPathExtractor`). LangChain’s `LLMGraphTransformer` (Chase, 2022; Bratanić, 2024) follows a similar pattern. In both modes users may supply `allowed_nodes` and `allowed_relationships` as schema constraints, with a strict flag that discards extractions violating the schema. The Neo4j LLM Knowledge Graph Builder (Neo4j Labs, 2024; Neo4j, 2024) offers three schema modes: a user-supplied schema of node types, relationship types, and valid source–relation–target triples; an auto-extracted schema inferred from the input text in a single upfront LLM call and held fixed thereafter; and a free mode with no schema.

These tools have seen wide adoption but share a fundamental design choice: the ontology — node types, relation types, and constraints — is specified arbitrarily and remains fixed for the duration of construction or even sometimes is built ad hoc by the LLM without an explicit schema.

Several academic approaches try to bridge triple extraction quality and task specification. Mo et al. (2025) propose KGGen, which extracts KGs from plain text using language models with improved entity and relation normalisation: an iterative clustering step collapses the near-duplicate relation types that open extraction produces, yielding denser and more connected graphs. Mynarz and Haniková (2023) propose a test-driven approach in which competency questions, formalised as SPARQL queries, drive extraction — providing a form of task specification without a quantitative feedback loop.

Li et al. (2025) provide the most comprehensive survey to date of LLMs across the ontology engineering lifecycle, reviewing 30 publications and 41 experiments spanning requirements specification, conceptualisation, encoding, ontology matching, and maintenance. The distribution of effort is stark: ontology implementation accounts for 63.4% of studied experiments, while ontology revision and maintenance has been addressed by a single study (2.4%). The survey names the consequences directly: “support across the ontology lifecycle is uneven, with maintenance particularly underexplored,” and finds that “evaluation practices are fragmented, as existing quantitative metrics fail to fully capture performance” (Li et al., 2025).

Across all reviewed studies, evaluation targets the ontology directly: the metrics in every case — precision, recall, F1 — are applied against ontology benchmarks such as the Ontology Alignment Evaluation Initiative (OAEI) rather than against a downstream application.

The survey closes by identifying ontology revision and maintenance as an open direction: “Future research directions may include developing hybrid learning pipelines that allow LLMs to continuously integrate domain updates ... thereby supporting ongoing validation and refinement based on domain-specific feedback” (Li et al., 2025).

The evaluation of knowledge graphs has historically been conducted through intrinsic,

structural quality metrics. Paulheim (2017) survey methods for KG completion and error detection, evaluating them against structural criteria such as precision and recall on missing type and relation assertions. More recently, Paulheim (2024) draw an explicit distinction between this intrinsic mode of evaluation and task-based evaluation — assessing a KG by its utility on a concrete downstream application. This distinction is central to the motivation of the present work: a graph may score well on structural criteria while failing to support a specific predictive task.

Several frameworks try to standardise task-based KG evaluation. KGBench Bloem et al. (2021) provides a suite of relational and multimodal machine learning tasks benchmarked on curated KG datasets. GEval Choi and Jung (2020) offers a software framework for evaluating KGs through downstream task performance across multiple dimensions. KGrEaT Heist et al. (2023) enables systematic comparison of KG variants across classification, regression, and recommendation tasks; the framework explicitly supports comparison of “different versions of a KG during its life cycle,” precisely the kind of comparison our feedback loop performs automatically. LLM-KG-Bench ? focuses specifically on the quality of LLM-generated graphs. Together, these frameworks confirm that task utility cannot be inferred from structural properties alone — yet none of them feeds performance signals back into the construction process itself.

The closest prior work to our approach concerns task-aware graph construction. Tsang et al. (2025) introduce AutoGraph-R1, the first framework to directly optimise KG construction for task performance using reinforcement learning: an LLM constructor is trained as a policy whose reward derives from the graph’s functional utility in a downstream RAG retrieval pipeline. AutoGraph-R1 shares with our approach the core principle of aligning construction with a downstream objective; the critical differences are that its reward derives from retrieval performance in a RAG pipeline rather than a node-level classifier, the RL policy optimises extraction density and triple relevance within an open-schema setting rather than selecting among ontology configurations, and there is no mechanism for explicit schema evolution.

Cucumides and Geerts (2025) propose constructing graphs from tabular and relational data to maximise GNN performance on a target task, tailoring graph topology through iterative process where each attribute’s predictive relevance is assessed. However, their setting involves structured tabular input rather than unstructured text, and the construction is not driven by an evolving ontology.

The financial domain chosen for this study provides a natural testbed for task-conditioned KG construction. Financial news significantly influences market dynamics (Dong et al., 2024), and a long line of work has explored how to represent its informational content as structured graphs for downstream prediction. ? demonstrate that event-driven representations of financial news — structured as subject–predicate–object triples capturing who did what to whom — yield strong signals for short- and medium-term stock price move-

ment when combined with deep neural networks. Subsequent work confirms and extends this finding: Ding et al. (2016) show that enriching event representations with knowledge graph embeddings further improves prediction accuracy, and attention-based models over news sequences (Hu et al., 2018) and financial social media text (Xu and Cohen, 2018) demonstrate the breadth of the predictive signal across sources and markets.

The key design question such approaches leave open is how to define the event schema: which entity types and relation types should be extracted from news to maximise predictive utility.

Motivated by the predictive utility of structured financial representations (Ding et al., 2016), domain-specific KG construction frameworks have also emerged. Arun et al. (2025) introduce FinReflectKG, a pipeline for constructing financial KGs from SEC 10-K filings using a pre-configured schema of entity and relationship types specific to financial regulatory documents. An LLM-based reflection component iterates to correct extraction errors within this fixed schema until no issues are detected or a step budget is exhausted.

Taken together, the literature reveals a consistent gap. Evaluation frameworks confirm that KG utility is task-dependent and cannot be inferred from structural properties alone (Heist et al., 2023; Bloem et al., 2021). Task-aware construction methods demonstrate that graph structure can be aligned with downstream objectives (Cucumides and Geerts, 2025; Tsang et al., 2025), and ? show that event-driven representations of financial news carry strong predictive signals for market classification (Ding et al., 2016; Hu et al., 2018; Xu and Cohen, 2018). Yet no existing work closes the loop: temporal KG methods evolve facts within a fixed schema (?); task-aware methods adapt topology or extraction reward without revising the ontology (Cucumides and Geerts, 2025; Tsang et al., 2025); financial KG systems such as FinReflectKG (Arun et al., 2025) iterate within a pre-configured schema; and ontology engineering surveys identify schema maintenance as the most underexplored stage of the lifecycle (Li et al., 2025). No existing method treats the KG ontology — the choice of node types, relation types, and constraints — as a hyperparameter to be optimised through a closed feedback loop between a downstream classifier and the construction process. The present paper addresses this gap directly.

3 Research Hypotheses

The proposed framework rests on the premise that knowledge graph ontology is not a neutral design choice but an active determinant of downstream predictive performance. If this is true, then iteratively adapting the ontology in response to task feedback should yield measurable improvements — both in accuracy and in the structural properties of the resulting graph. We formalise this premise through three testable hypotheses, each targeting a distinct dimension of the expected benefit: predictive accuracy, graph efficiency, and structural informativeness.

Hypothesis 1. *Task-aware knowledge graph construction improves prediction accuracy in node-level price movement classification compared to static graph construction approaches.*

Ontology configurations that are better aligned with the prediction objective should produce graph-derived features with higher discriminative power. H1 is evaluated by comparing AUC and F1-score of the downstream classifier trained on features extracted from task-aware KG variants against a static baseline in which the ontology is fixed throughout (Section 6).

Hypothesis 2. *Task-aware optimization enables smaller knowledge graphs while maintaining or improving downstream task performance.*

By discarding ontology elements that do not contribute to predictive performance, the feedback loop should converge on more compact representations. H2 is evaluated by tracking the number of nodes and edges per iteration alongside classifier AUC, testing whether the task-aware variant achieves a favourable size–performance tradeoff relative to the unconstrained baseline (Section 6).

Hypothesis 3. *Iterative, task-driven knowledge graph construction produces more informative graph structures and facilitates the identification of effective ontology configurations.*

Successive ontology revisions guided by performance feedback should increase the diversity and predictive relevance of extracted relation types. H3 is evaluated by examining the ontology evolution trace — which relation types are added, retained, or removed across iterations — and by measuring feature entropy and metapath diversity over successive construction steps (Section 6).

4 Methodology

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A complementary body of work addresses how to derive informative features from knowledge graphs for downstream prediction. Selected methods operate on heterogeneous information networks (HINs) — networks whose nodes and edges carry type labels defined by an explicit network schema (?). Metapath-based approaches Sun et al. (2011); Dong et al. (2017) exploit typed paths in such networks to capture relational semantics between node pairs. Knowledge graph embedding methods survey a broad class of techniques for representing entities and relations in continuous space Wang et al. (2017); Ji et al. (2021), supporting link prediction and node-level classification. Temporal extensions of this family, such as Know-Evolve (?), model facts and entity representations evolving over time within a fixed schema.

All these methods share the assumption that the HIN schema — the set of node types and relation types — is given and fixed; feature engineering, embedding, or message-passing then operates over the resulting structure. Even temporal extensions model what changes within a fixed schema, not how the schema itself should be designed. In our framework we draw on metapath counting and relation type histograms as components of the feature vector, but the network schema from which these features are derived is itself adapted in response to the downstream task. Rather than learning better representations from a fixed graph, we ask what schema produces the most predictively useful structure.

5 Experimental Setup

[TODO: Write this section]

6 Results

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7 Discussion

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8 Conclusion

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